BIHAR ENGINEERING UNIVERSITY, PATNA

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

B.Tech 4th Semester • Examination Resource

ANALOG ELECTRONICS & INTEGRATED CIRCUITS

Exam Date: Saturday, 31 Oct 2026 (2:00 PM - 5:00 PM)

Included Question Papers

Year	Paper Title	Course Code	Examination Session
2024	Analog Circuits	104401	B.Tech 4th Semester Examination, 2024
2023	ECE Analog Circuit	104401	End Semester Examination - 2023 Semester-IV
2022	Analog Electronic Circuits	110401	B.Tech 4th Semester Exam., 2022
2020	Linear Integrated Circuits and Applications	104504	B.Tech 5th Semester Exam., 2020 (New Course)

Apna BEU Question Bank Archive • Compiled for B.Tech ECE 4th Semester

Contains official previous year questions with multiple-choice questions, descriptive problems, and complete circuits.

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 4th Semester Examination, 2024

ANALOG CIRCUITS

Time: 03 Hours | Code: 104401 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Choose the correct option/answer the following (Any seven question only): **[14]**

1. JFET can be operated with
   - i. negative gate only
   - ii. gate voltage only
   - iii. positive as well as negative gate voltage
   - iv. none of the above

2. Which power amplifier has the highest collector efficiency
   - i. C
   - ii. A
   - iii. B
   - iv. AB

3. Which amplifier is used for 180-degree phase shift
   - i. differential amplifier
   - ii. non inverting amplifier
   - iii. inverting amplifier
   - iv. None of the above

4. A MOSFET has three terminals, namely
   - i. cathode, anode, grid
   - ii. source, gate, drain
   - iii. emitter, base, collector
   - iv. none of the above

5. The maximum efficiency of resistance loaded class B power amplifier is
   - i. 5%
   - ii. 50%
   - iii. 30%
   - iv. None of the above

6. A JFET is also called _____ transistor
   - i. unipolar
   - ii. bipolar
   - iii. injunction
   - iv. none of the above

7. In a phase shift oscillator, the frequency determining elements are
   - i. L and C
   - ii. R, L and C
   - iii. R and C
   - iv. None of the above

8. The leakage current across a pn junction is due to
   - i. Minority carriers
   - ii. Majority carriers
   - iii. Junction capacitance
   - iv. None of the above

9. An oscillator converts
   - i. a.c. power into d.c. power
   - ii. a.c. power into a.c. power
   - iii. mechanical power into a.c. power
   - iv. none of the above

10. In a transistor
    - i. $I_E = I_C - I_B$
    - ii. $I_B = I_C + I_E$
    - iii. $I_C = I_E + I_B$
    - iv. $I_E = I_C + I_B$

Q.2 Solve both questions : **[14]**

1. In a full wave rectifier, the input is from 30-0-30V transformer. The load and diode forward resistances are 100Ω and 10 Ω respectively. Calculate the average voltage, dc output power, ac input power, rectification efficiency and percentage regulation.

2. Explain the working of positive clamping circuit.

Q.3 Solve both questions : **[14]**

1. Draw the common emitter circuit and sketch the input and output characteristics. Also explain active region, cutoff region and saturation region by indicating them on the characteristic curve.

2. With a neat circuit diagram explain the Voltage Divider Bias circuit by giving its exact analysis.

Q.4 Solve both questions : **[14]**

1. Explain the characteristics of an Ideal Op-Amp. Mention some of the applications of Op-Amp.

2. Explain R-2R ladder DAC with example?

Q.5 Solve both questions : **[14]**

1. Explain Successive type ADC with neat circuit diagram.

2. Describe the construction and operation of JFET and draw its characteristics.

Q.6 Solve both questions : **[14]**

1. Explain Colpitts oscillator with Circuit diagram and derive its frequency.

2. Explain RC coupled transistor amplifier with neat circuit diagram in a middle frequency Range.

Q.7 Solve both questions : **[14]**

1. Explain RC- phase shift oscillator using op-amp and derive its frequency.

2. Describe the construction and operation of depletion type MOSFET and draw its characteristics.

Q.8 Solve both questions : **[14]**

1. Describe class A Power Amplifier. Derive the expression for its maximum conversion Efficiency.

2. Explain Wien bridge oscillator using op-amp and derive its frequency.

Q.9 Solve both questions : **[14]**

1. Explain MOSFET as an amplifier?

2. Describe class B Power Amplifier. Derive the expression for its maximum conversion Efficiency.

BIHAR ENGINEERING UNIVERSITY, PATNA

End Semester Examination - 2023 Semester-IV

ECE ANALOG CIRCUIT

Time: 03 Hours | Code: 104401 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Write the answer of the following (Any seven question only): **[14]**

1. Which of these are incorrect about Darlington amplifier?
   i. It has a high input resistance
   ii. The output resistance is low
   iii. It has a unity voltage gain
   iv. It is a current buffer

2. What is a cascade amplifier?
   i. A cascade of two CE amplifiers
   ii. A cascade of two CB amplifiers
   iii. A cascade of CE and CB amplifiers
   iv. A cascade of CB and CC amplifiers

3. What is the output of a class B amplifier for sinusoidal input?
   i. Sinusoidal amplifier
   ii. Half-sinusoidal
   iii. Sinusoidal with higher frequency
   iv. Square wave

4. In a class B amplifier, it is found that DC power is 25W, find the ac power.
   i. 10 W
   ii. 62.5 W
   iii. 25 W
   iv. 50 W

5. In an oscillator if phase of feedback is same as that of oscillation waveform then feedback is called
   i. Positive feedback
   ii. Negative feedback
   iii. Cannot be predicted
   iv. Either positive or negative depending upon frequency

6. For a MOSFET, the pinch-off voltage is -3V. Gate to source voltage is 20V. W/L ratio is 5. Process transconductance parameter is 40$\mu A/V^2$. Find drain to source current in saturation.
   i. 0.10 mA
   ii. 0.05mA
   iii. - 0.05mA
   iv. -50A

7. For a MOSFET, the pinch-off voltage is -3V. Gate to source voltage is 20V. W/L ratio is 5. Process transconductance parameter is 40$\mu A/V^2$. Find drain to source current in saturation.
   i. 1100 Ω
   ii. 11000 Ω
   iii. 900 Ω
   iv. 110 Ω

8. The hybrid π model is used for
   i. Analysis of BJT for low frequency
   ii. Analysis of BJT at high frequency
   iii. Analysis of BJT for feedback
   iv. Analysis of BJT for large signal

9. Which oscillator is characterized by a split capacitor in its tank circuit?
   i. RC phase shift oscillator
   ii. Colpitts oscillator
   iii. Wien bridge oscillator
   iv. none

10. The gate voltage in a JFET at which drain current becomes zero is called
    i. pinch-off voltage.
    ii. saturation voltage
    iii. active voltage.
    iv. cut-off voltage.

Q.2 Solve this question : **[14]**

1. Explain construction and working of Enhancement and Depletion type MOSFET.

Q.3 Solve this question : **[14]**

1. What is power amplifier. Explain Transformer coupled Class-A amplifier.

Q.4 Solve this question : **[14]**

1. Barkhausen criteria play an important role while designing Oscillator justify. Explain Hartley and RC phase shift Oscillator.

Q.5 Solve this question : **[14]**

1. Explain the effect of voltage series and voltage shunt feedback on gain & impedance of amplifier.

Q.6 Solve this question : **[14]**

1. The BJT is connected as a CE amplifier. Determine Z_i, Z_0, A_v, A_i hybrid model.

Q.7 Solve this question : **[14]**

1. Describe current mirror circuit in BJT. What are the condition to design current mirror circuit. Obtain the expression for current transfer ratio.

Q.8 Solve this question : **[14]**

1. Derive the expression for voltage gain and current gain and B.W of multistage amplifier.

Q.9 Solve this question : **[14]**

1. In brief explain the working of voltage series and current series circuit. Explain the same in case of BJT.

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 4th Semester Exam., 2022

ANALOG ELECTRONIC CIRCUITS

Time: 03 Hours | Code: 110401 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Attempt any seven from the following: **[14]**

a. What is avalanche multiplication?

b. Compare CE, CC and CB for current gain, voltage gain, input and output impedance.

c. Junction transistor is an active device. Justify.

d. Write the equation of a diode and draw its VI characteristics.

e. What is early effect in bipolar junction transistors?

f. Draw an npn transistor in CB configuration and draw its input characteristics.

g. BJT has higher driving capability as compared to MOSFET. Justify.

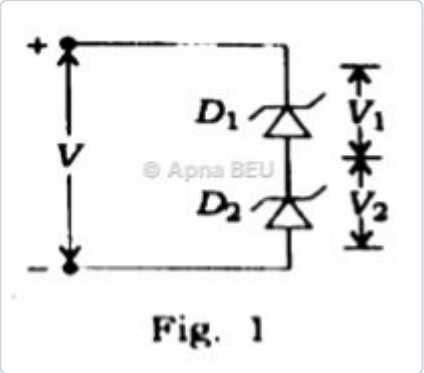
h. For an ideal negative feedback configuration, there is a virtual short circuit between the input terminals. Why?

i. Draw a circuit which can be used as limiter or clamper or both.

Q.2 Solve both questions : **[14]**

1. Write brief notes on: (i) Avalanche breakdown (ii) Zener breakdown

2. The saturation currents of the diodes are 1 μA and 2 μA. The breakdown voltages of the diodes are the same and are equal to 100 V: (i) Calculate the current and voltage for each diode if V = 90 V and V = 110 V (ii) Repeat part (i) if each diode is shunted by 10 MΩ resistor.



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Fig. 1

Q.3 Solve both questions : **[14]**

1. With reference to rectifying circuits, explain- (i) ripple factor (γ); (ii) transformer utility factor (TUF); (iii) peak inverse voltage (PIV); (iv) rectifier efficiency (η).

2. For HW rectifier and capacitor filter- (i) draw the circuit diagram; (ii) draw the waveforms; (iii) give a brief explanation of the functioning of the circuit.

Q.4 Solve both questions : **[14]**

1. Draw Ebers Moll model for pnp transistor and show that transistor action is not possible by connecting two diodes back to back.

2. For the transistor shown below, $\alpha = 0.98$ and $V_{BB} = 0.7$, find R_1 for an emitter current $I_E = -2mA$. Neglect I_{CO}.

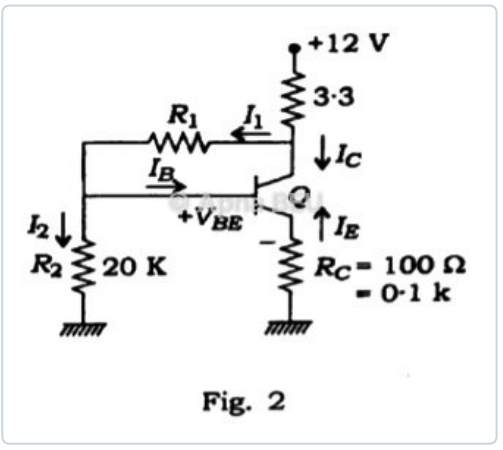


Fig. 2

Q.5 Solve both questions : **[14]**

1. Given the following measurements made at $I_C = 5mA$, $V_{CE} = 10V$ and at room temperature $h_{fe} = 100$, $h_{ie} = 600\Omega$, $A_{ie} = 10$ at 10 MHz, $C_C = 3pF$. Find f_T, f_α, $r_{b'e}$ and $r_{bb'}$ for the transistor.

2. A silicon transistor with $\beta = h_{FE} = 100$, $V_{BE}(sat) = 0.8V$, $V_{CE}(sat) = 0.2V$ is used in the circuit shown. Find the minimum value of R_C for which transistor remains in saturation.

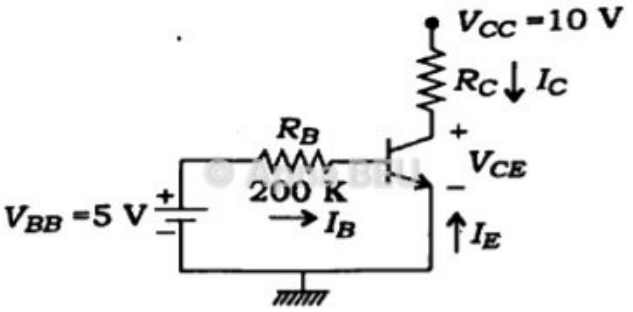


Fig. 3

Q.6 Solve both questions : **[14]**

1. Compare the performance of MOSFET and BJT.

2. Explain the construction of p-channel MOSFET. Draw and explain both depletion and enhancement mode MOSFET.

Q.7 Solve both questions : **[14]**

1. Calculate v_0 and the op-amp's output current as shown in the circuit and if E_i equals (i) +5 V and (ii) -2 V. For each situation, state if the op-amp sources or sinks current.

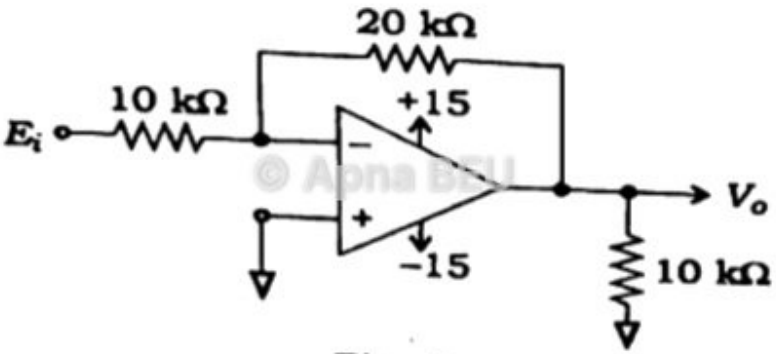
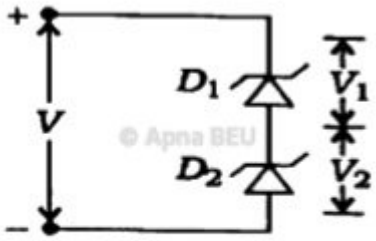


Fig. 4

2. (i) Design an inverting amplifier with a gain of -5 and an input resistance of 10 kΩ. (ii) Design a non-inverting amplifier with a gain of 5.

Q.8 Solve both questions : **[14]**

1. The circuit shown is required to supply on average voltage of 120 V. Determine the r.m.s. value of secondary voltage, average current delivered to the load, peak current through each diode and PIV of diodes.



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Fig. 5

2. Explain different amplifiers with suitable example.

Q.9 Write short notes on the following: **[14]**

a. Precision comparator

b. Zero crossing detector with hysteresis

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 5th Semester Exam., 2020 (New Course)

LINEAR INTEGRATED CIRCUITS AND APPLICATIONS

Time: 03 Hours | Code: 104504 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are EIGHT questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Answer any seven of the following short answer-type questions : **[2×7=14]**

a. Write two requirements to be met for a good current source.

b. Why VCO is also called as v-to-f converter?

c. What is a switched capacitor filter?

d. What output voltage would be produced by a D/A converter whose output range is 0 to 10 V and whose input binary number is 0110 for a 4-bit DAC?

e. What is a power amplifier?

f. What is the purpose of connecting a capacitor at the input and the output side of an IC voltage regulator?

g. Mention any two applications of PLL.

h. For an n-bit flash-type A/D converter, how many comparators are required?

i. What is meant by frequency synthesizing?

j. Why are integrators preferred over differentiators?

Q.2 Solve both questions : **[7+7=14]**

a. With suitable circuit diagram, explain the operation of voltage reference circuit using temperature compensation.

b. What are the different methods used in fabrication of integrated resistors? Explain any one method with suitable diagrams.

Q.3 Solve both questions : **[7+7=14]**

a. Draw and explain the circuit of a voltage to current converter if the load is (i) floating and (ii) grounded.

b. Define and explain slew rate. What is full-power bandwidth? Also explain the methods to improve the slew rate.

Q.4 Solve both questions : **[7+7=14]**

a. Explain the working of a anti logarithmic amplifier using suitable op-amp circuit diagram and supporting equations.

b. Draw the functional diagram of 723 regulators and explain its working.

Q.5 Solve both questions : **[7+7=14]**

a. Draw the block diagram and explain the principle of working, characteristic and applications of a frequency shift keying (FSK) demodulator.

b. Explain the operation of frequency to voltage converter with suitable circuit diagram.

Q.6 Solve both questions : **[7+7=14]**

a. Explain the working of R-2R ladder D/A converter with circuit diagram.

b. Explain the oversampling A/D converter with functional block diagram.

Q.7 Solve both questions : **[7+7=14]**

a. Explain with block diagram how PLL is used as frequency modulation (FM) detector.

b. Design a VCO having the maximum range of 1 kHz. Assume power supply $V_{cc} = 15\ \mathrm{V}$. Required pulse width should not exceed 100 $\mu\mathrm{s}$.

Q.8 Solve both questions : **[7+7=14]**

a. Design an astable multivibrator circuit which will flash the electric bulb such that its on time will be 4 seconds and off time will be 2 seconds.

b. Explain the working of Schmitt trigger using 555 timer.